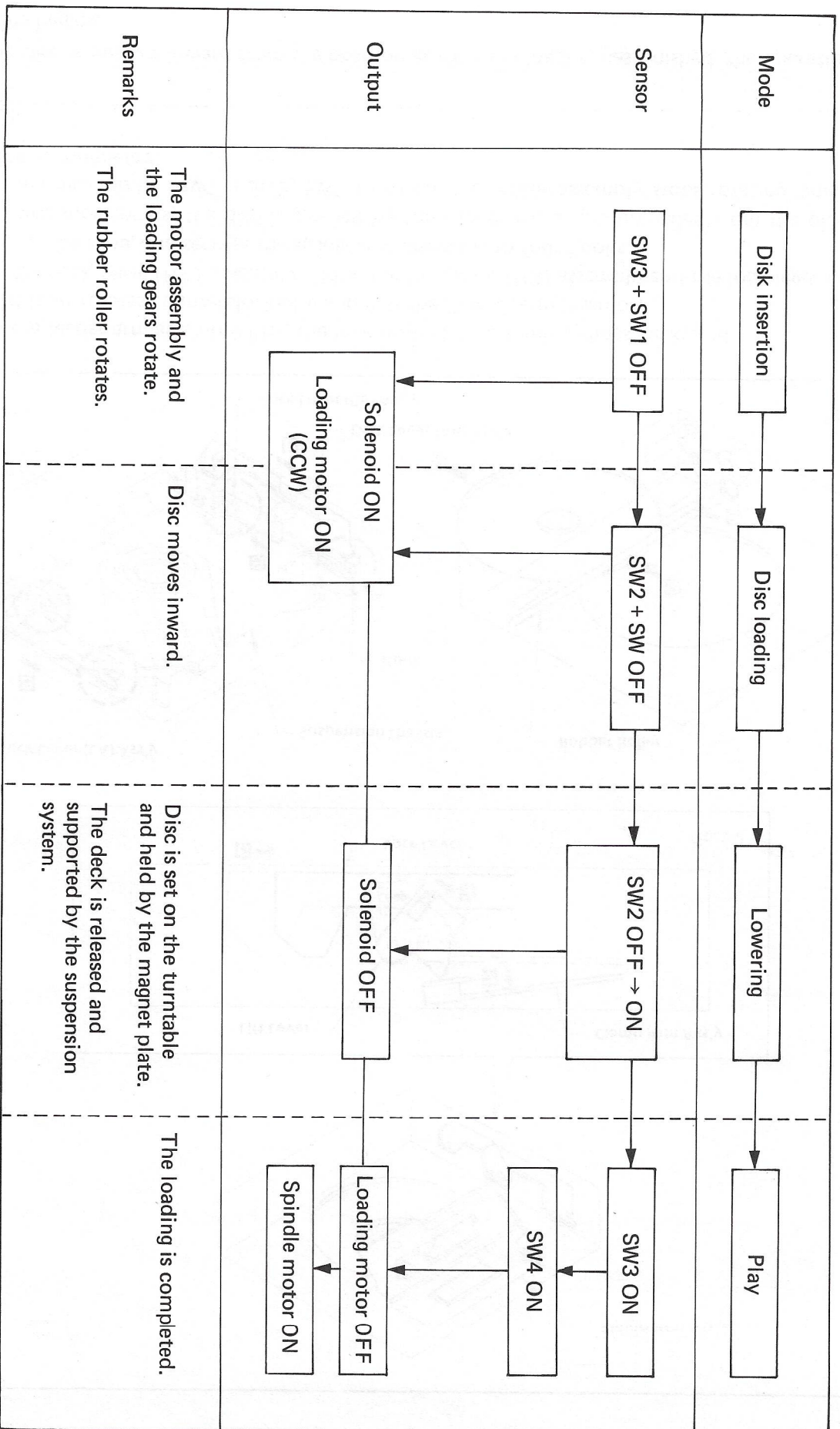


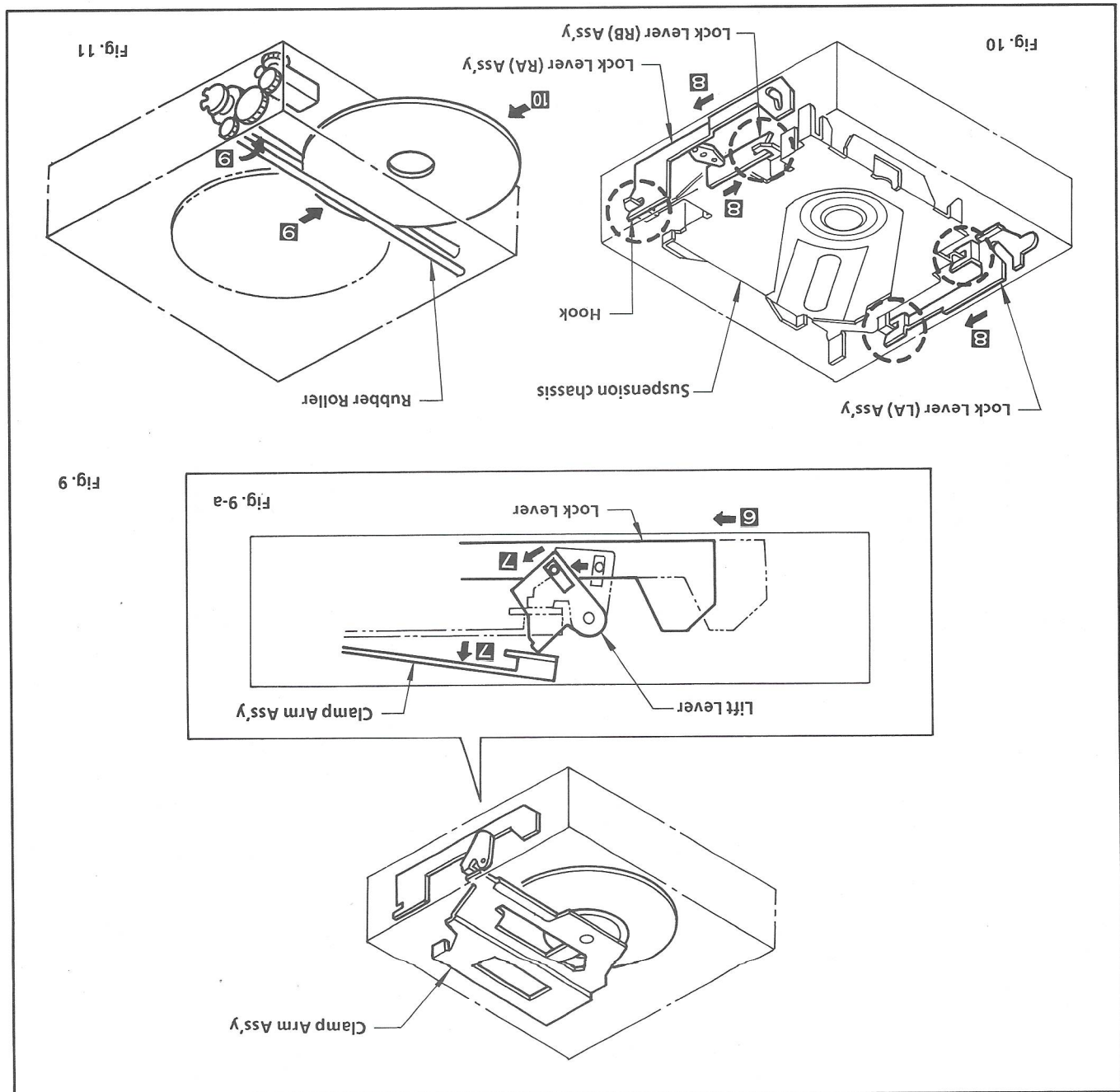
3-4 Loading Operation



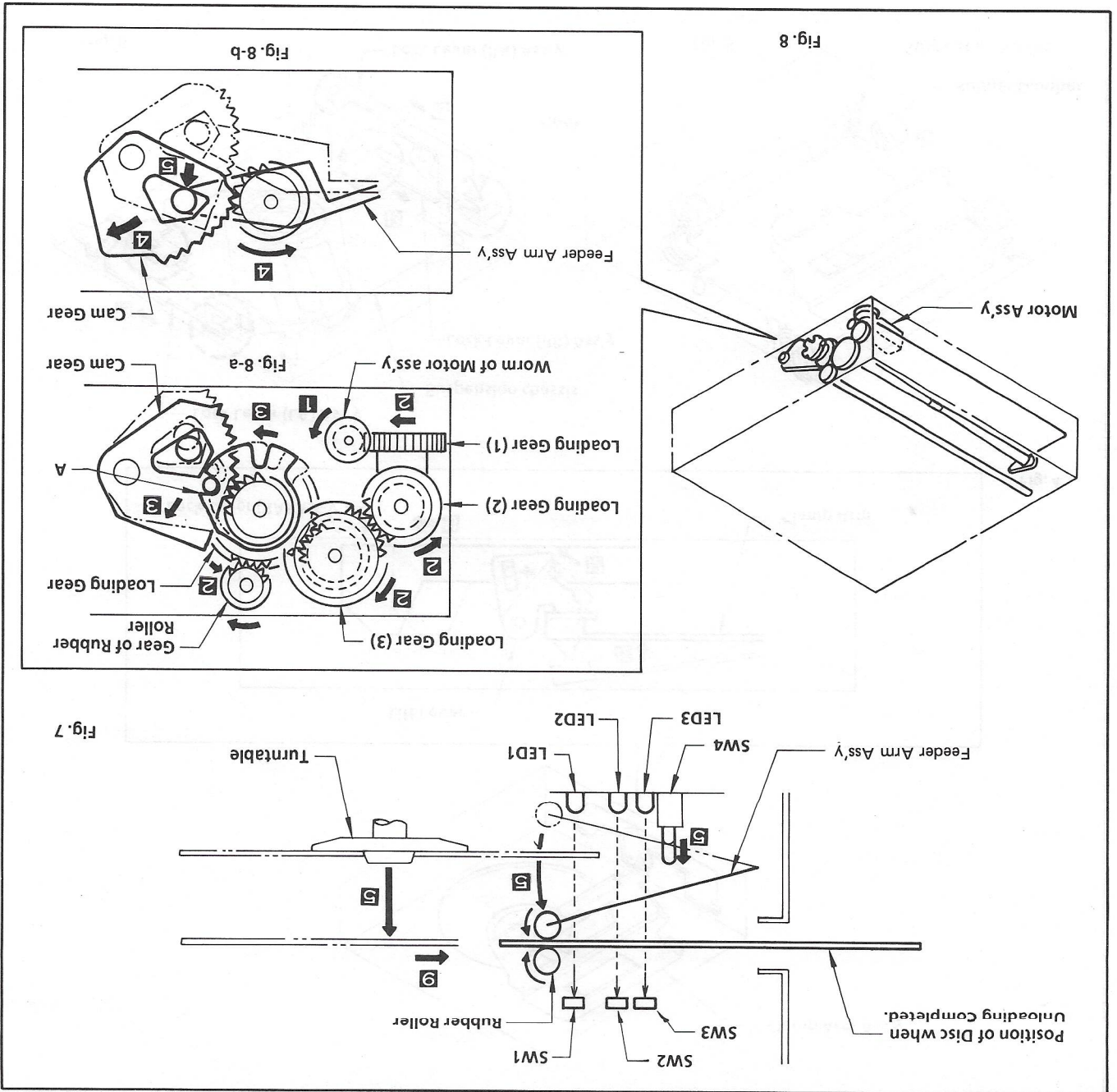
Reloading

10 If the disc is pushed inward from the position at which unloading has finished, the operation of loading begins.

- 6 When the feeder arm assembly lifts, the lock lever (RA) assembly slides backward.
- 7 The lift lever rotates counterclockwise and lifts the Clamp Arm assembly.
- 8 When the lock lever (RA) assembly slides, the lock lever (RB) assembly and the lock lever (LA) assembly also slide, and secures the suspension chassis with four hooks.
- 9 With these movements, the disc is grasped by the rubber roller and unloaded from the player. Ejection is completed.



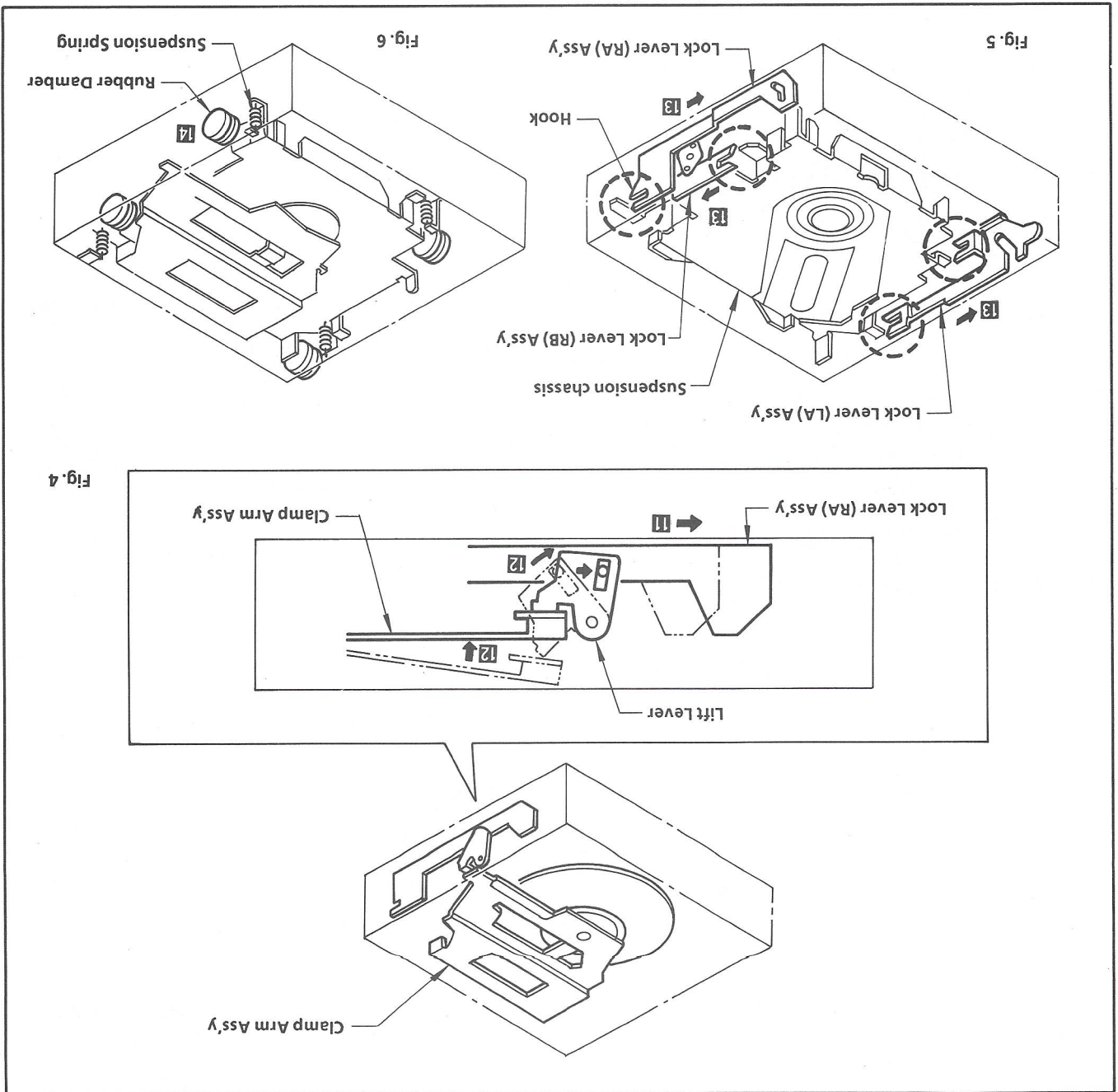
A3-3 Unloading/Reloading Operation



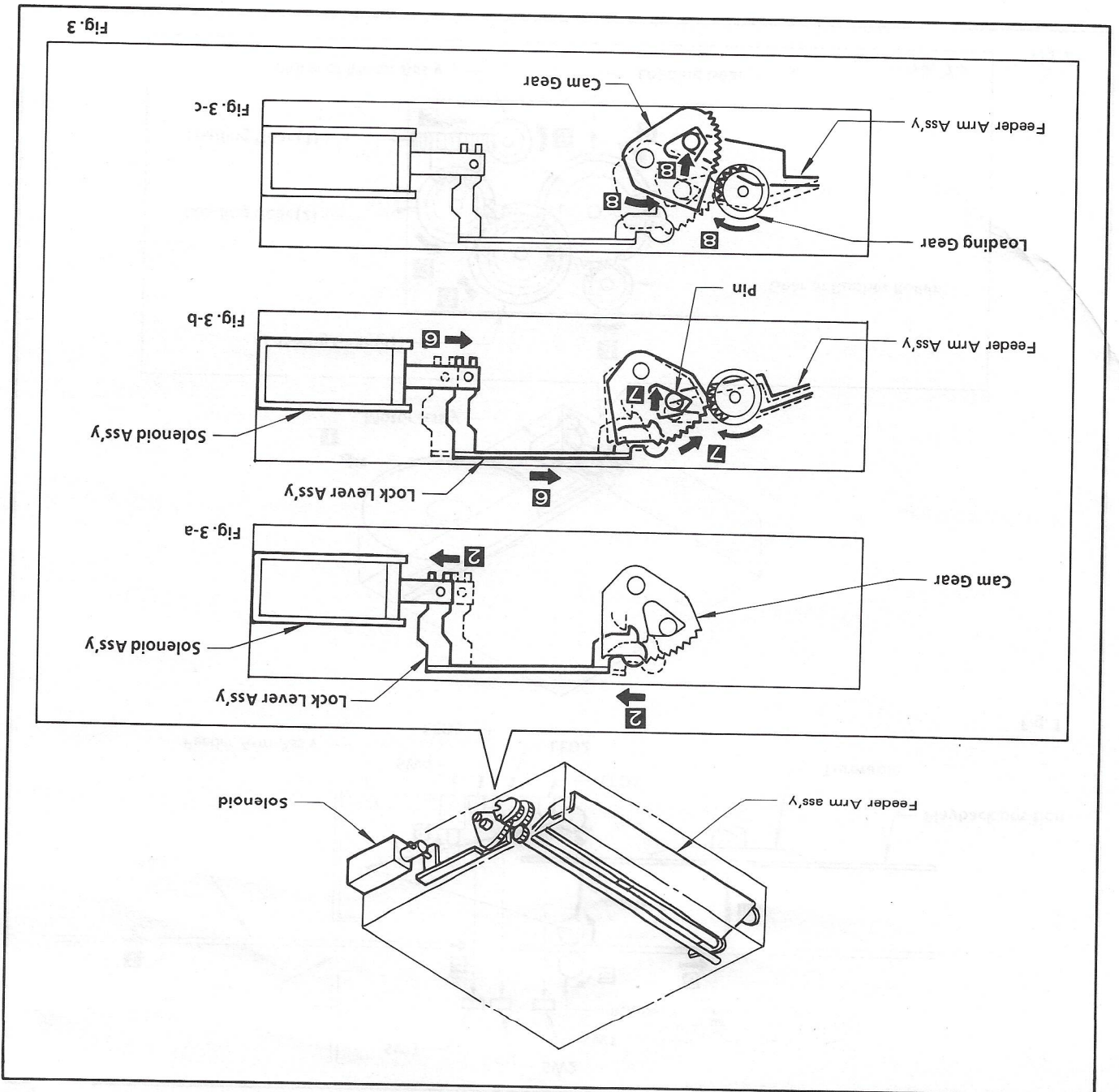
Unloading

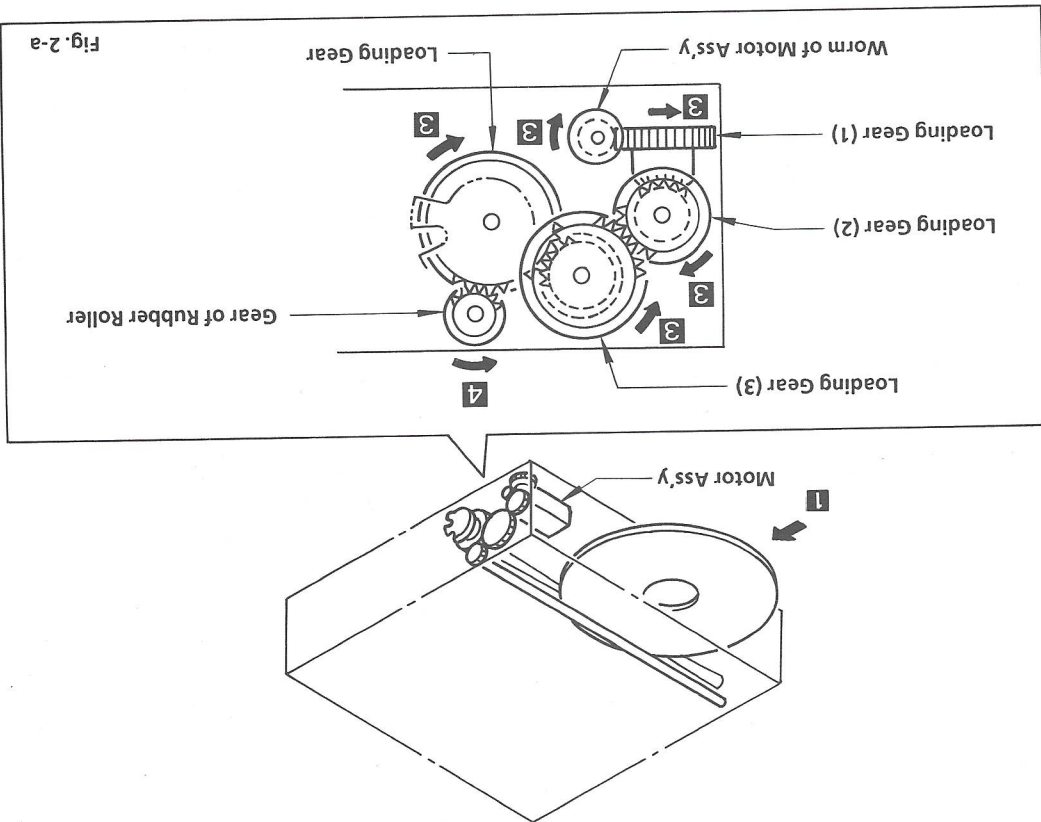
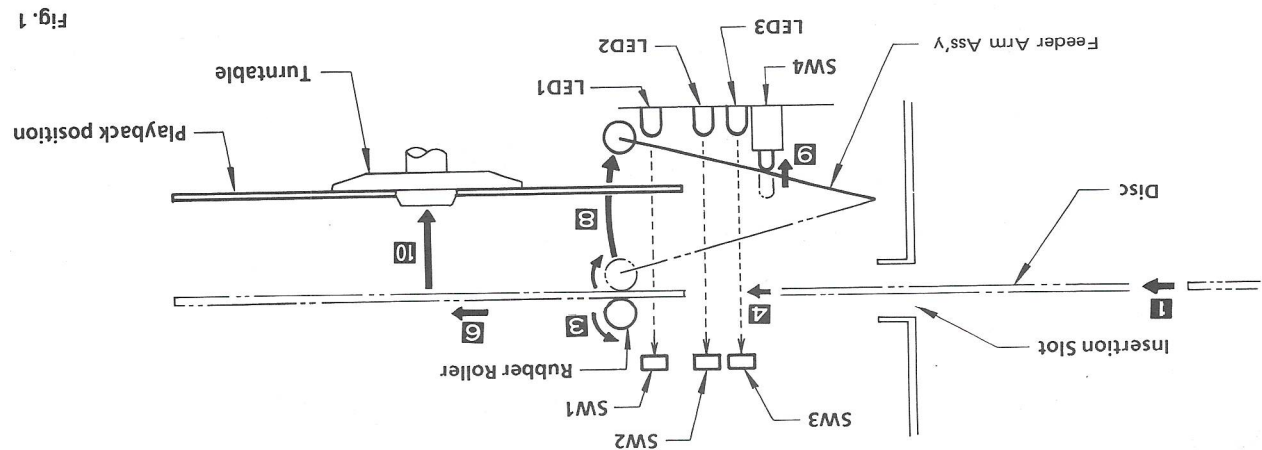
- 1 When the eject signal is received, the motor assembly rotates in the direction opposite that of loading.
- 2 The loading gears rotate.
- 3 Part A on the loading gear rotates the cam gear slightly clockwise, then the cam gear and loading gear are engaged.
- 4 The loading gear rotates the cam gear clockwise.
- 5 The cam gear lifts the feeder arm assembly. (SW4 : Off)

- 11 When the feeder arm assembly lowers, the lock lever (RA) assembly slides forward.
- 12 The lift lever rotates clockwise and lowers the clamp arm assembly.
- 13 The disc is secured in the position for playback by the force of the magnet.
- 14 When the lock lever (RA) assembly slides forward, the lock lever (RB) assembly and the lock lever (LA) assembly also slide, and the secured suspension chassis are released.
- 14 The suspension chassis is then supported by the suspension springs and rubber dampers.



- 6 When the disc moves beyond SW2 (SW2: ON), the solenoid is turned off, and the hook is released.
- 7 The weight of the disc lowers the feeder arm assembly slightly. Its lowering force rotates the cam gear slightly counterclockwise, then the cam gear and loading gear are engaged.
- 8 The loading gear rotates clockwise and the cam gear rotates counterclockwise.
- 9 The rotation of the cam gear lowers the feeder arm assembly and turns on SW4.
- 10 The disc also lowers and is set on the turntable of optical pickup.



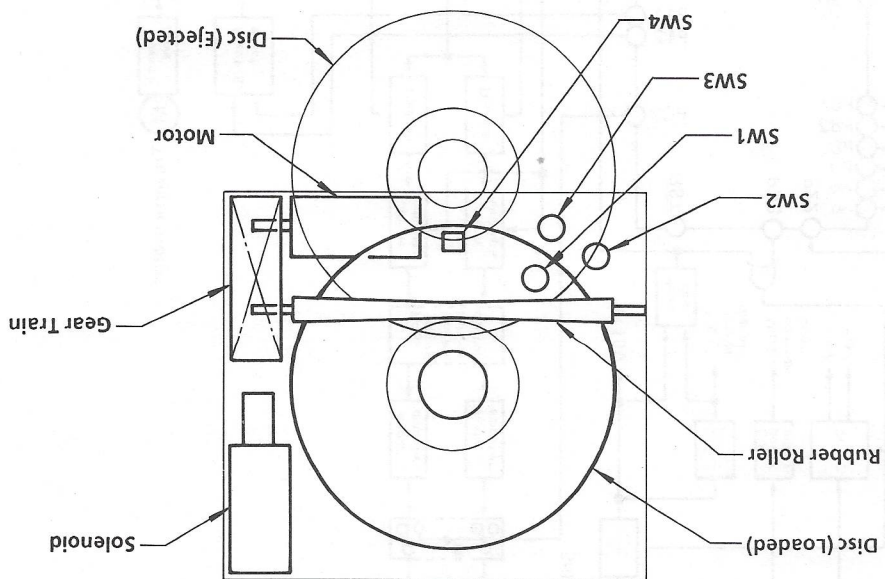


Loading

- 1 Insert the disc into the player.
 - 2 When SW3 turns OFF, the solenoid is activated the lock lever assembly slides, and its hook secures the cam gear.
 - 3 SW1 turns OFF. The motor assembly and the loading gears rotate.
 - 4 The rubber roller rotates.
 - 5 The disc will grasp by the two rubber rollers and load into the player.
- * SW1, SW2 and SW3 are ON when No Disc.

3. Loading Mechanism And It's Operation (TYPE A)

A3-1 Disk Loading Mechanism



Name	Function	Description	Remarks
SW1	Disk in detection	To detect whether or not a disk is inserted.	Photensor
SW2	Disk position detection	Detects the timing for Switching from horizontal movement to vertical movement. Reference point for determining how far to eject the disk.	Photensor
SW3	Disk loaded or unloaded detection	To detect whether the disk has been properly loaded. To detect whether the disk has been properly unloaded.	Photensor
SW4	Mechanism status detection	To detect whether the disk loading mechanism is in play mode.	Contact Switch (Micro Switch)
MOTOR	Disk loading motor	Motor for loading and unloading the disk.	Direction of motor Loading: CCW Unloading: CW
SOL	Clutch Switching	Switches disk movement from vertical to horizontal.	When solenoid on Horizontal movement
Rubber Roller	To move Disk in and out	Converts motor rotary motion into liner motion.	
Gear train			

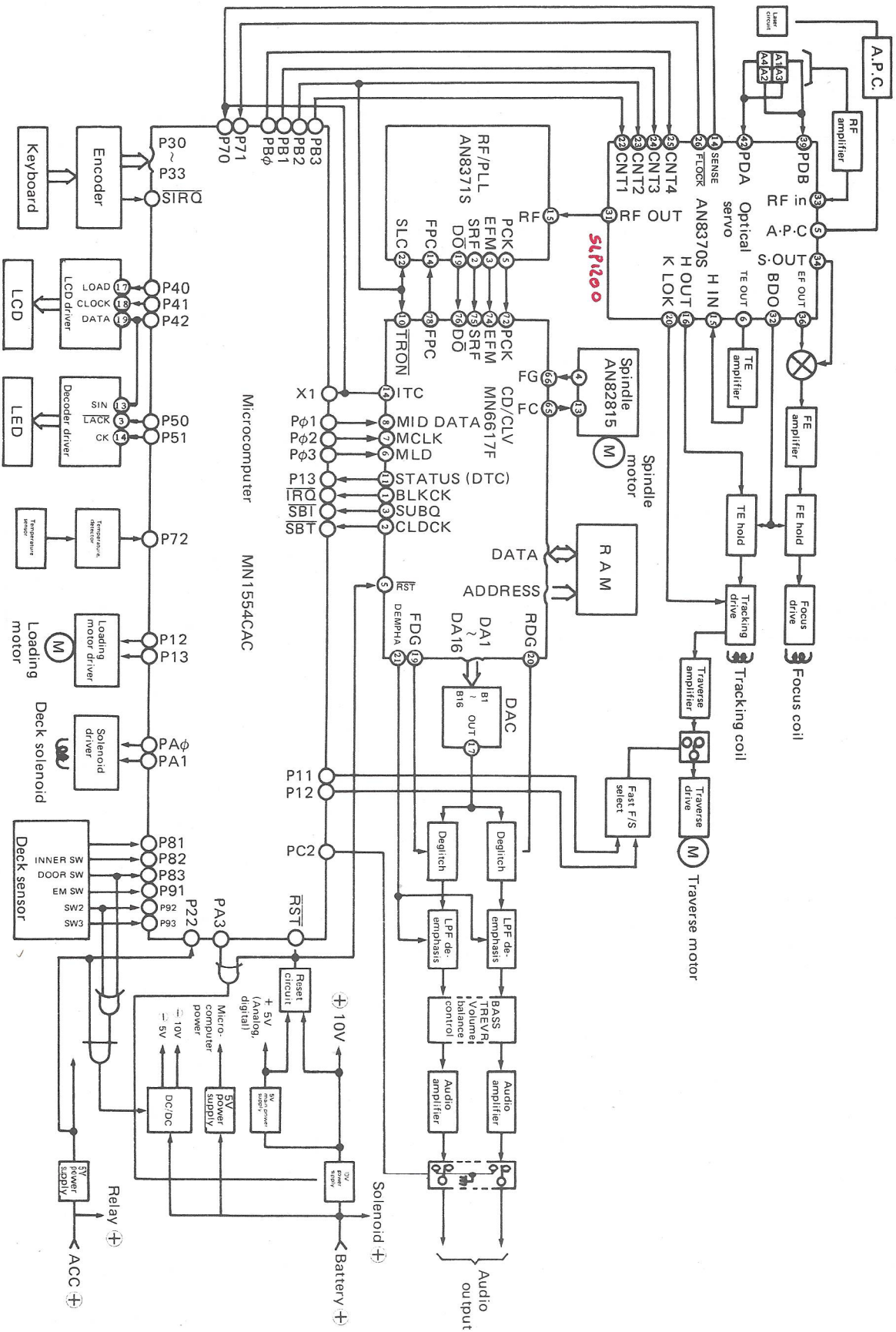
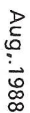


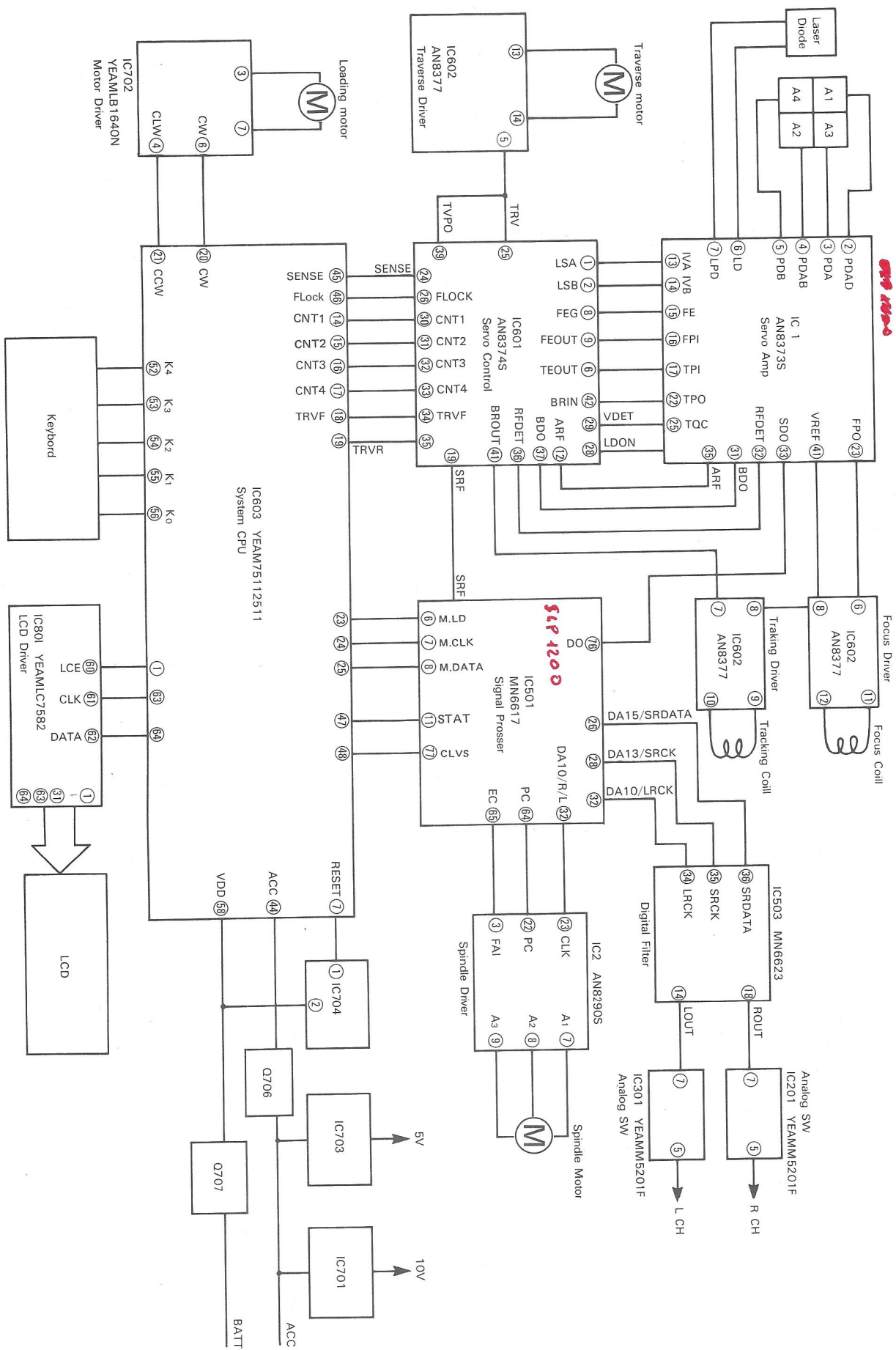
Fig. 2-1 Block Diagram

- 9 -



2. SYSTEM BLOCK DIAGRAM

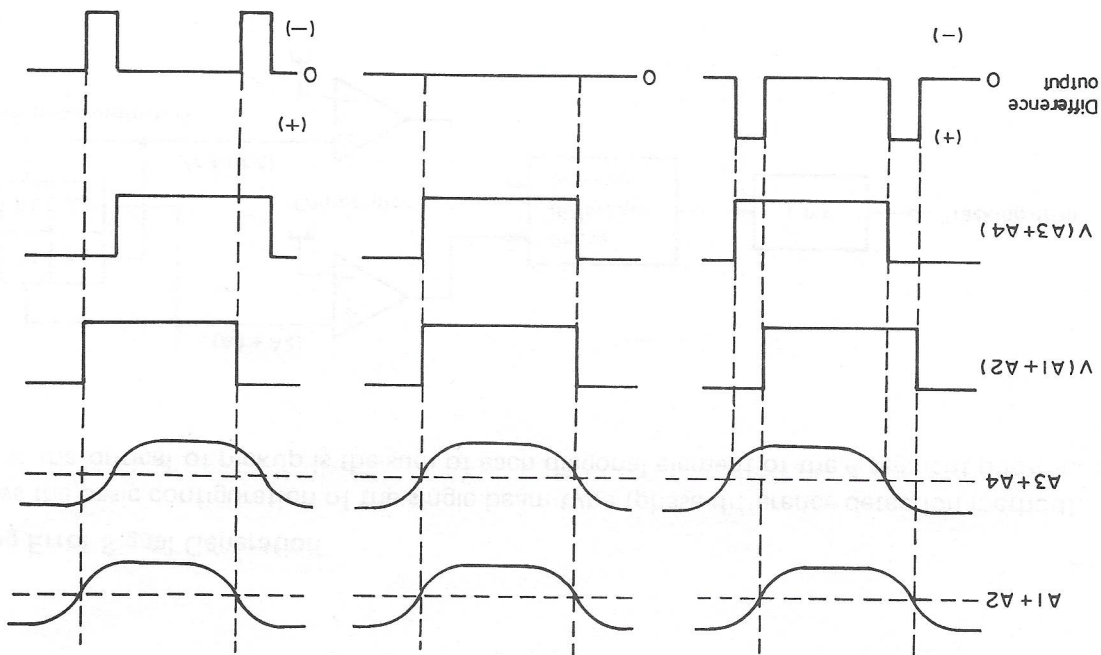
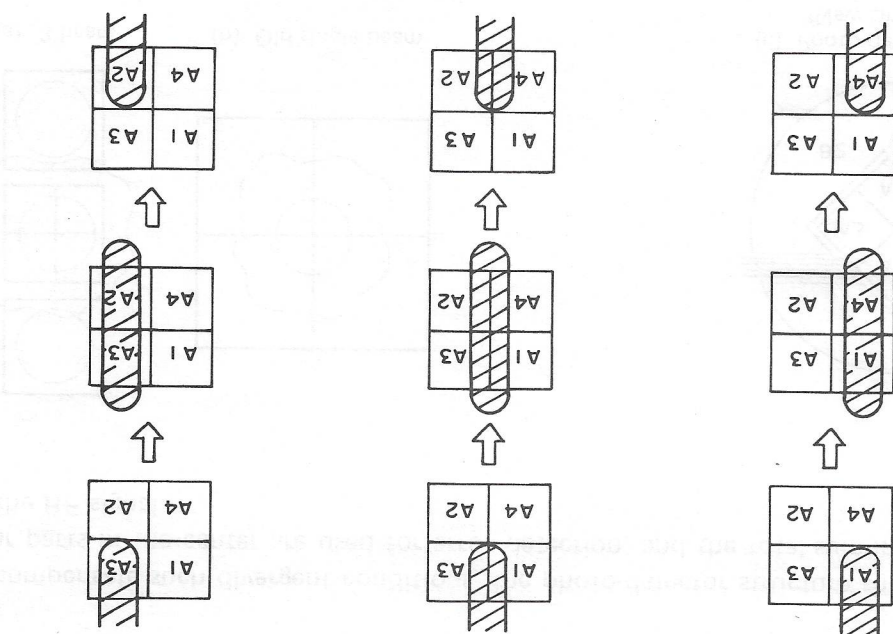
2-1 SYSTEM BLOCK DIAGRAM (Type A)



If a track pit image runs over the 4-segment photo-detector and deviates to the left or the right, a phase difference corresponding to the track deviation is produced between the output signals ($A1 + A2$) and ($A3 + A4$). This phase difference is detected by the phase detector, and passed through the lowpass filter (L.P.F.) to generate a tracking error signal.

Fig. 1-5 shows a tracking error detection diagram.

(a) Track Deviating to the Left (b) Track in the Center (c) Track Deviating to the Right



● A positive pulse is generated if $V(A1 + A2)$ leads $V(A3 + A4)$, and a negative pulse is generated if vice versa.

Fig. 1-5 Tracking Error Detection Diagram

1-3 Photo Detector

The photo detector is probably the single most important component in the compact disc player. The reason for this is that all intelligence needed to operate the unit is developed from the returning laser beam by the photo detector. This will include focus error signal, tracking error signal as well as the intelligence signal.

The single beam type (Fig. 1-3b) has a large PD surface area to prevent RF signal level reduction. However, if the beam irradiated at a scratch or other defects on the disc surface, the servo errors occur and decrease playability.

In order to compensate such divergent conditions, the photo-detector structure shown in Figure 1-3c is used, the four parts in the center are used for error detection, and the total sum including the periphery ($B1 + B2$) is the RF signal.

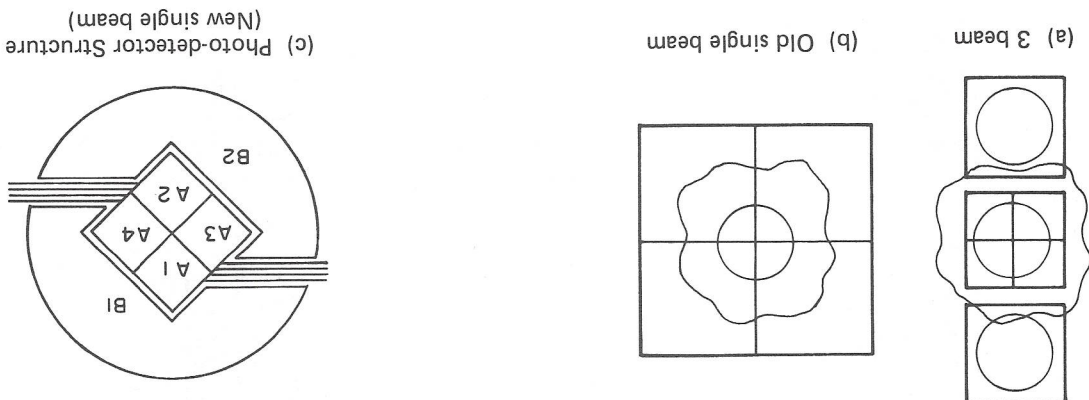


Fig. 1-3

1-4 Tracking Error Signal Generation

Fig. 1-4 shows the basic configuration of the single beam type (phase difference detection method). The output of the optical pickup is the sum of each diagonal element of the 4-segment photo-detector.

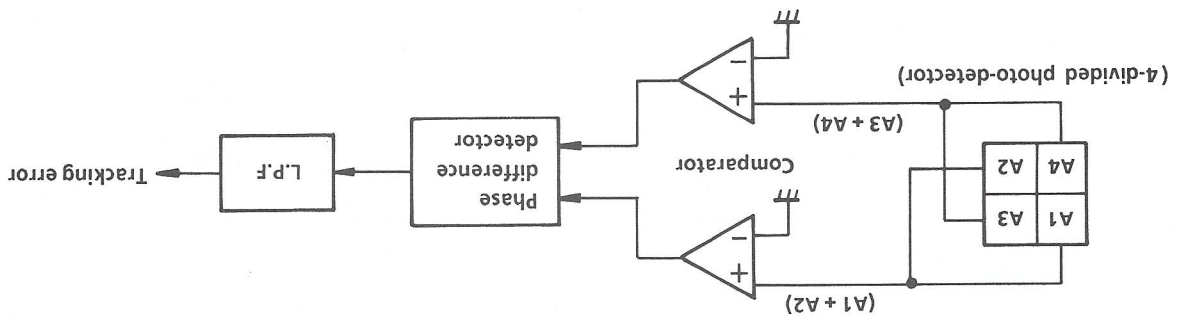


Fig. 1-4 Basic Configuration

- 1-2 3-Beam Type vs. Single Beam Type
- 1-2-1 The largest difference in optical structure between the 3-beam type and single beam type is that single beam type has no diffraction grating. Therefore, the single beam type has only one beam on optical path, and its photo-detector is divided into four instead of six (four plus two).
- 1-2-2 The two sub-beams of the 3-beam type have the function of generating tracking errors signal. The process of generating tracking error signal by the single beam type that is without sub-beams is described in Section 1-4. Focus error signal generated by the conventional astigma method.
- 1-2-3 3-Beam type has a small RF detector area ($150\text{ }\mu\text{m}$ squared). Therefore, it has a disadvantage when playing discs with fingerprints.
- 1-2-4 1-Beam type (4-segments) overcomes the fingerprint playability but it has a disadvantage when playing scratched discs. The area of the 4-segment photodetector is large ($1000\text{ }\mu\text{m}$ squared). Therefore, the error generated by fingerprints is included and corrected. Since the focus and tracking detectors are included as part of the RF detector, the spike is detected as an error signal.
- 1-2-5 1-Beam type (6-segments) has a reduced size RF detector ($250\text{ }\mu\text{m}$ squared) and two large and separate detectors have been added. Any spikes generated by a scratched disc do not appear as a large spikes to the tracking and focus detectors. The total area of the photodetector is now $1400\text{ }\mu\text{m}$ squared.

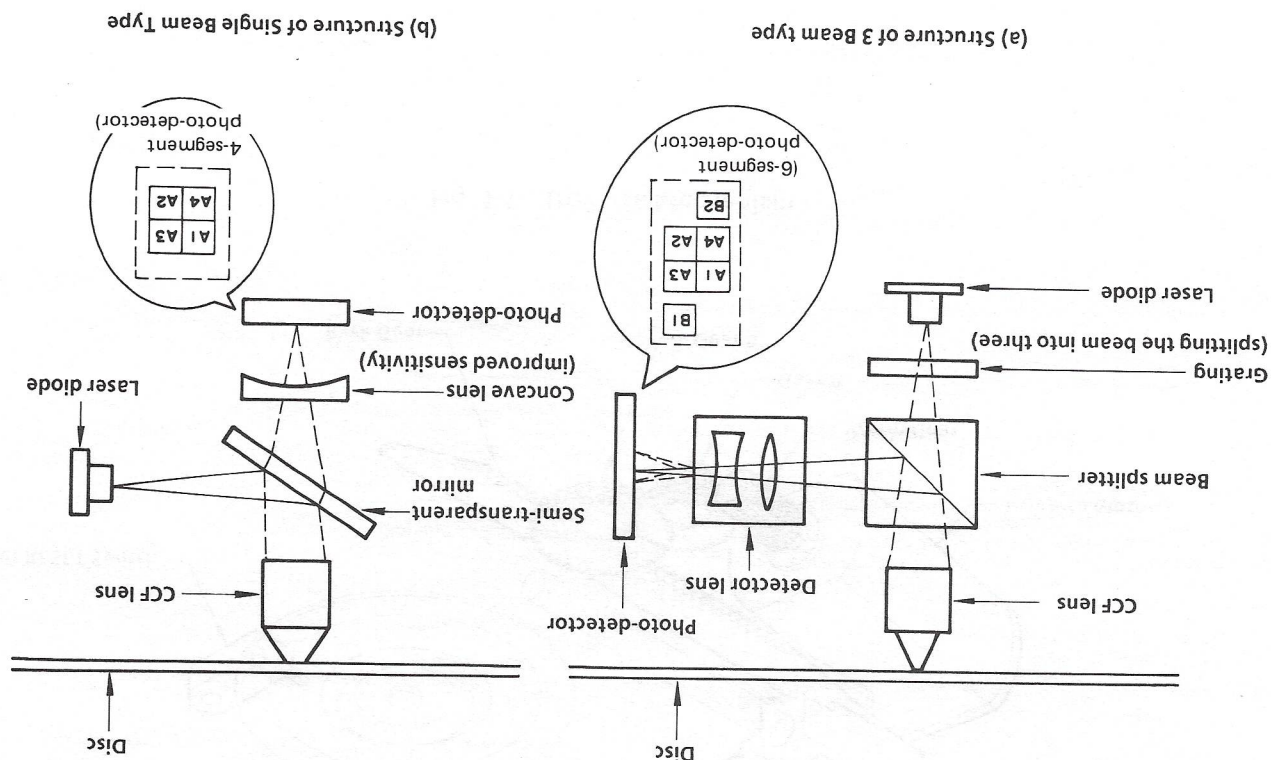


Fig. 1-2 Structure of Optical Pickup

1. Traverse Mechanism And Optical Pick Up

1-1 Traverse Mechanism

The traverse mechanism (Fig. 1-1) drives the optical pickup base by the rack gear which is driven by the motor via the worm gear, gear A, and gear B. The gear A and gear B are double clutch gears. The inner RESET switch is turned on by the optical pickup base when the optical pickup is driven innermost position by the traverse mechanism.

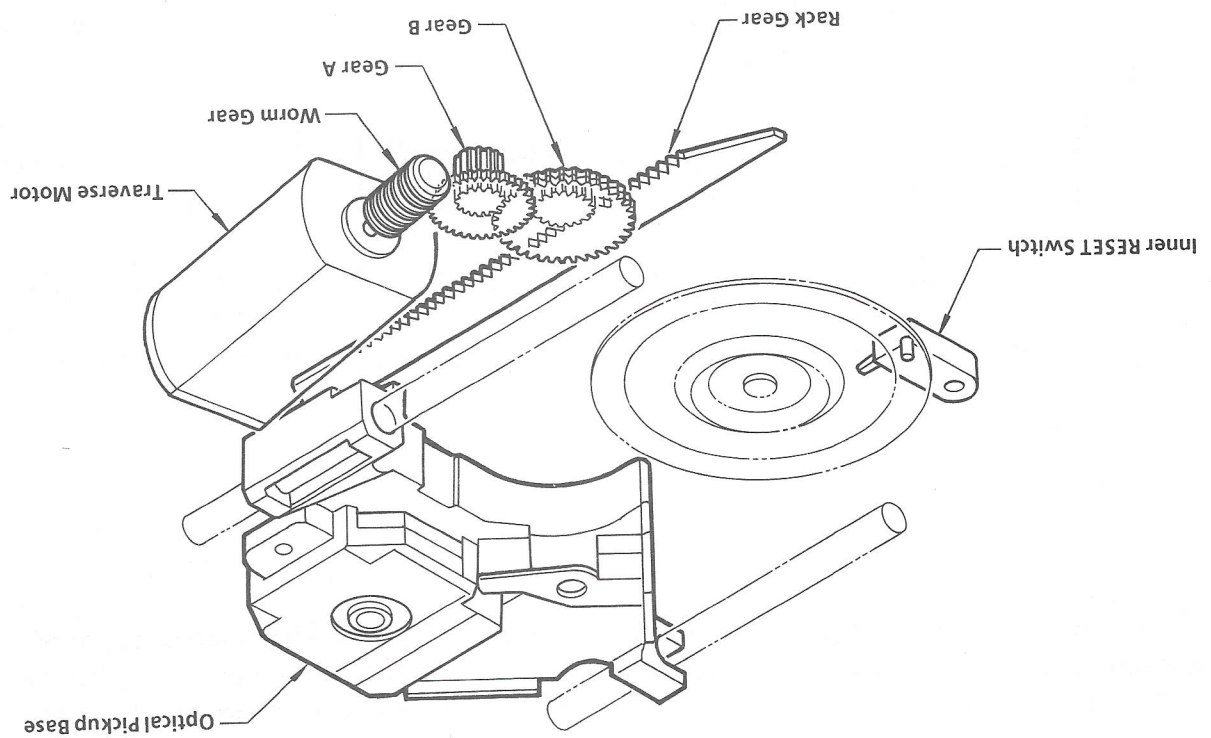


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	CO-DP25EC	CO-DP26EG
A	CX-DP11EU/EC	CX-DP11EN
	CX-DP10EU/EC	CX-DP10EN
B	CO-DP5EU/EC	CO-DP5EG
	CO-E800EU	CX-DP3EN
C		

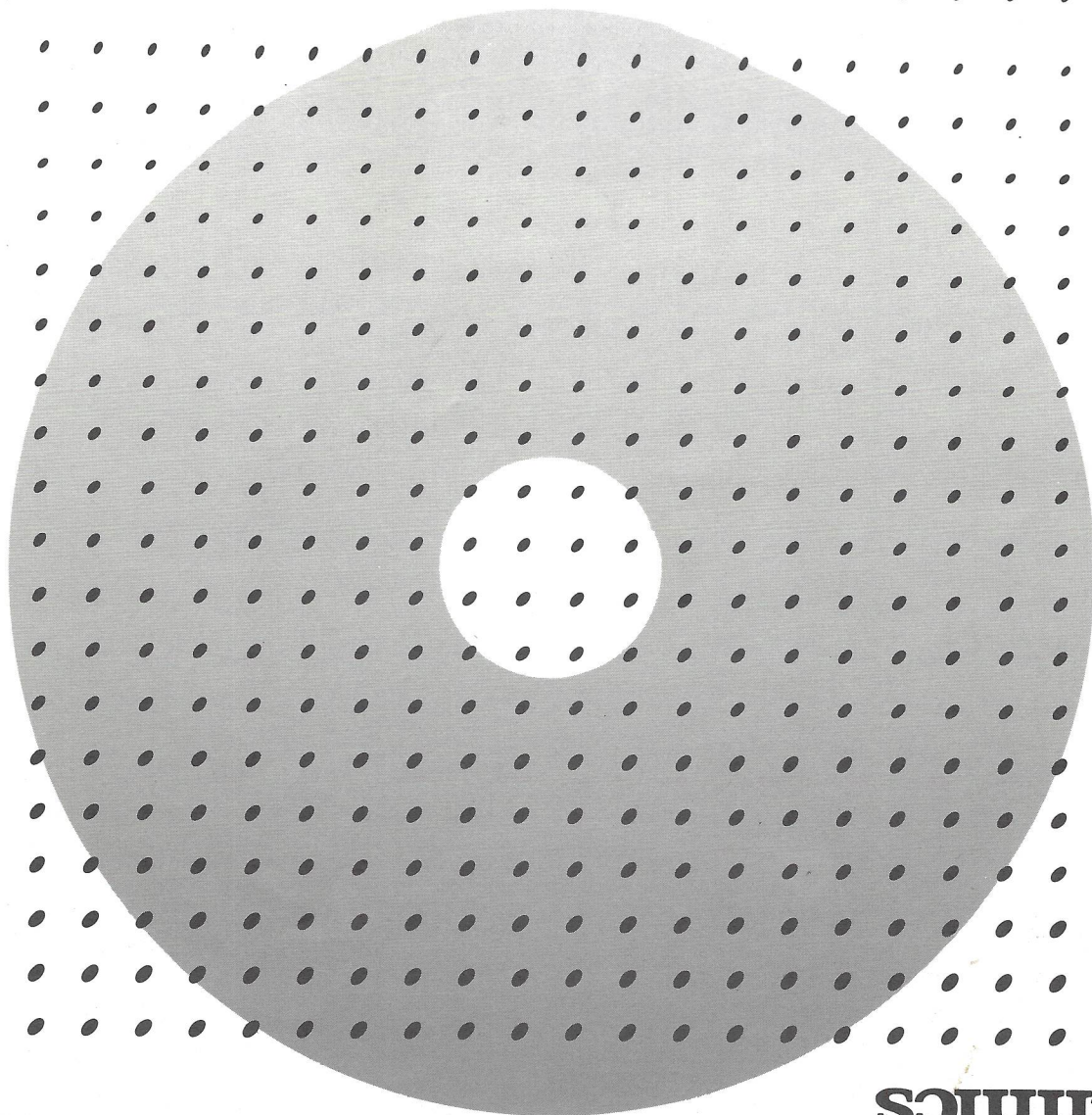
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